Calculus I

Computation problems #2

Topics:

- Definition of limits
- Infinite limits
- Continuity

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Problem 1. Prove by definition that $\lim\limits_{x\to 4}(9-x)=5$.

Problem 2. Prove by definition that $\lim\limits_{x\to -3}\dfrac{x^2-9}{x+3}=-6$.

Problem 3. Find $\lim\left(\dfrac{x^2}{2}-\dfrac{1}{x}\right)$ as

a) $x\to 0^+$; **b)** $x\to 0^-$; **c)** $x\to \sqrt[3]{2}$; **d)** $x\to -1$.

Problem 4. Find $\lim\limits_{t\to 0^+}\left(2-\dfrac{3}{t^{1/3}}\right)$ and $\lim\limits_{t\to 0^-}\left(2-\dfrac{3}{t^{1/3}}\right)$.

Problem 5. Prove by definition that $\lim\limits_{x\to 0}\dfrac{-1}{x^2}=-\infty$.

Problem 6. Prove by definition that $\lim\limits_{x\to -5}\dfrac{1}{(x+5)^2}=+\infty$.

Problem 7. Prove by definition that $\lim\limits_{x\to 2^-}\dfrac{1}{x-2}=-\infty$.

Problem 8. At what points are the function $y=\dfrac{\cos x}{x}$ continuous?

Problem 9. At what points are the function $y=(2x-1)^{1/3}$ continuous?

Problem 10. Prove by definition that the function $h(x)=\sqrt{2x-3}$ is continuous at $x_0=2$.

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Answer to selected problems

3. (a) $-\infty$ (b) $+\infty$ (c) 0 (d) $\dfrac{3}{2}$

4 . (a) $-\infty$ (b) $+\infty$

8. $\mathbb{R} \setminus \{0\}$

9. $\mathbb{R}$